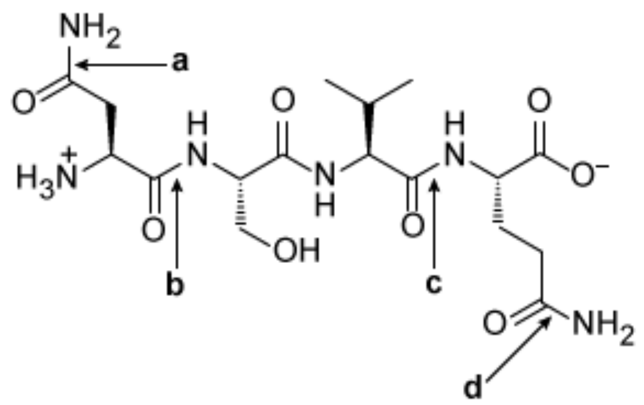
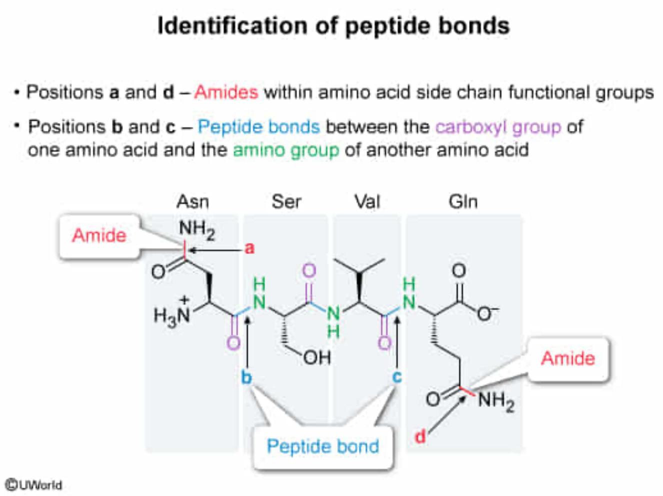


Which bond(s) in the peptide shown above CANNOT be classified as a peptide bond?

- A. Position **a** only [1%]
- B. Positions **a** and **d** only [89%]
- C. Positions **b** and **c** only [7%]
- D. Positions **a**, **c**, and **d** only [1%]

Correct
90% answered correctly

Explanation:



Amino acids are organic molecules containing a carboxyl group, an amino group, and a side chain, all bonded to the same carbon atom (the α-carbon). Peptides are made of two or more amino acids linked together via **peptide bonds**, which are a type of amide linkage. Peptide bonds form when the **carboxyl group** of one amino acid residue condenses with the **amino group** of another amino acid residue via a dehydration reaction.

This question shows a peptide with four bonds labeled. The peptide comprises four amino acids (from the amino terminus to carboxyl terminus): asparagine, serine, valine, and glutamine. The amides linking amino acid residues are peptide bonds; positions **b** and **c** identify the peptide bonds that link asparagine to serine and valine to glutamine, respectively. The amino acid residues at the amino and carboxyl termini (asparagine and glutamine, respectively) have side chains that contain amides (positions **a** and **d**).

The question asks which of the labeled bonds *cannot* be classified as a peptide bond. Because the amides at **position a and position d** are within amino acid residue side chains, they **cannot be classified as peptide bonds**.

(Choice A) Although position **a** is not a peptide bond, it is not the *only* position that cannot be classified as a peptide bond. Position **d** is an amide within an amino acid side chain, and therefore also *not* a peptide bond.

(Choice C) Positions **b** and **c** both *can* be classified as peptide bonds. The question asks for which positions *cannot* be classified as peptide bonds.

(Choice D) Position **c** is an amide linking valine to glutamine, and therefore *is* a peptide bond.

Educational objective:
Peptides comprise two or more amino acids linked together through peptide bonds—a type of amide. A peptide bond is formed when the carboxyl group of one amino acid residue condenses with the amino group of another amino acid residue.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404109
System:
Functional Groups and Their Reactions